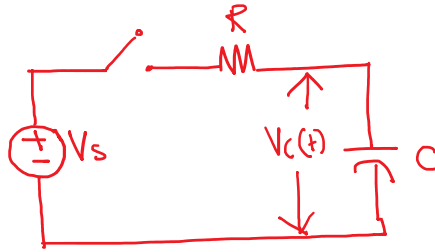
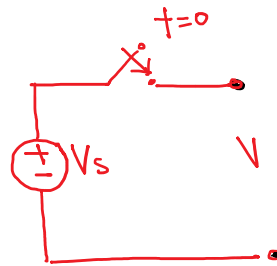
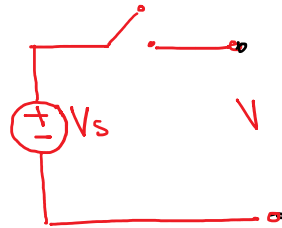


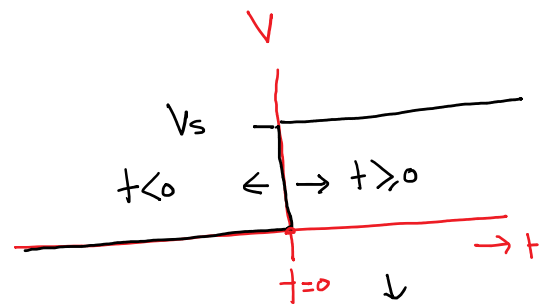
Forced RC ckt



$$t < 0: V = 0V$$



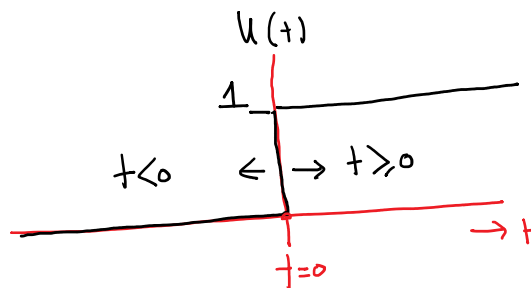
$$t \geq 0: V = V_s$$



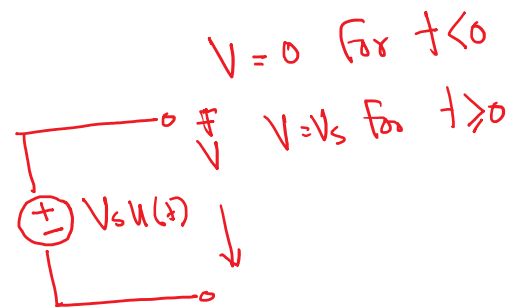
Unit Step function.

$$u(t) = 0 \text{ for } t < 0$$

$$u(t) = 1 \text{ for } t \geq 0$$

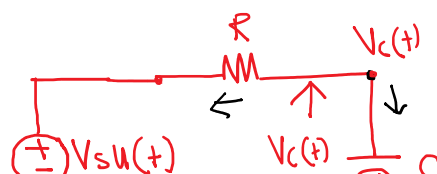


$$V_s u(t)$$



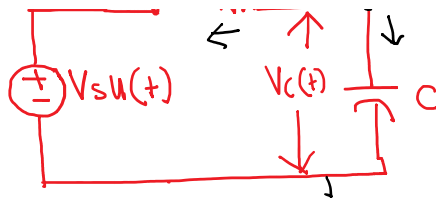
$$V = 0 \text{ for } t < 0$$

$$V = V_s \text{ for } t \geq 0$$



$$\frac{V_c(t) - V_s u(t)}{R} + C \frac{dV_c(t)}{dt} = 0$$

$$\dots \text{ or } dV_c(t)$$



$$\Rightarrow V_c(t) - V_s u(t) + R_c \frac{dV_c(t)}{dt} = 0$$

$$t \geq 0, \quad V_s u(t) = V_s \times 1 \quad \Rightarrow \quad V_c(t) - V_s + R_c \frac{dV_c(t)}{dt} = 0$$

$$\Rightarrow \frac{dV_c(t)}{dt} + \frac{V_c(t)}{R_c} = \frac{V_s}{R_c} \quad \varphi$$

$$\frac{dx}{dt} + Px = Q$$

$$x(t) = \frac{Q}{P} + Ae^{-Pt}$$

$$P = \frac{1}{R_c}$$

$$Q = \frac{V_s}{R_c}$$

$$\frac{dx}{dt} + Px = Q$$

$$x(t) = \frac{Q}{P} + Ae^{-Pt}$$

$$V_c(t) = \frac{\frac{V_s}{R_c}}{\frac{1}{R_c}} + Ae^{-\frac{t}{R_c}}$$

Forced R_c

$$V_c(t) = V_s + Ae^{-t/R_c}$$

$$\text{at } t=0, \quad V_c(0) = V_s + A e^{-0/R_c}$$

$$\Rightarrow A = V_c(0) - V_s$$

$$V_c(t) = V_s + [V_c(0) - V_s] e^{-t/R_c}$$

$$\text{at } t=\infty \quad V_c(\infty) = V_s + [V_c(0) - V_s] e^{-\infty/R_c}$$

$$V_s = V_c(\infty)$$

$$V_c(t) = V_c(\infty) + [V_c(0) - V_c(\infty)] e^{-t/R_c}$$

Need to determine 3 parameters

$$\tau = R_c$$

$$= R_{eq} C_{eq}$$

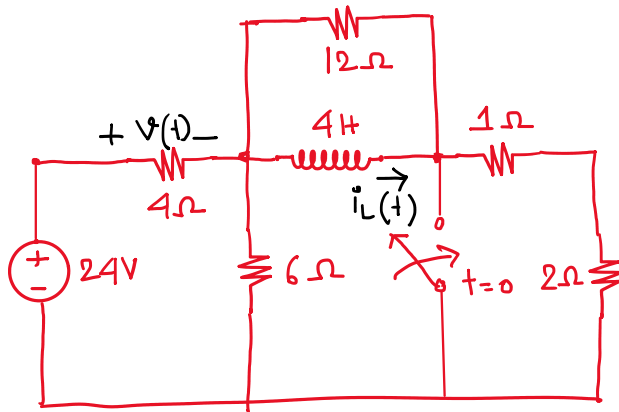
$$1. V_c(0) = V_c(0^-) \quad [t < 0] \rightarrow \text{CKT}$$

$$2. V_c(\infty) \quad [t > 0] \rightarrow \text{CKT}$$

$$3. R_{eq} = R_{TH} \quad [t > 0] \rightarrow \text{CKT}$$

$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)] e^{-t/L/R}$$

$$\tau = L/R \quad | \quad = \frac{L_{eq}}{R_{eq}}$$

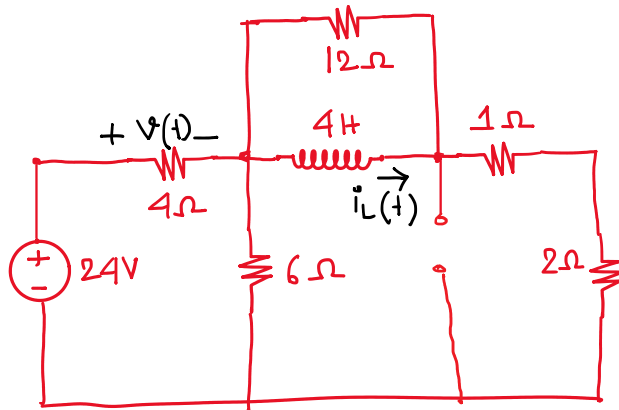


$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)] e^{-t/\tau}$$

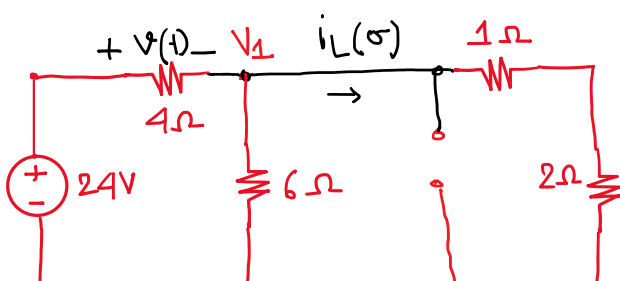
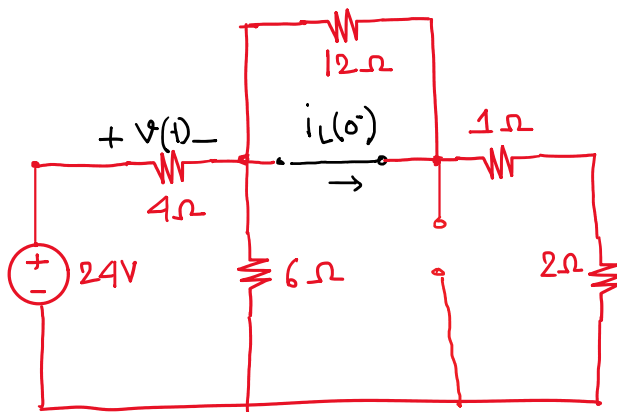
1. $i_L(0) : (t < 0)$
2. $i_L(\infty) : (t > 0)$
3. $\tau : (t > 0)$

$i_L(0) : \text{for } t < 0$. Switch open

$$i_L(0) = i_L(0^-)$$



Switch open for long time,
inductor s.c.k.s to DC signal

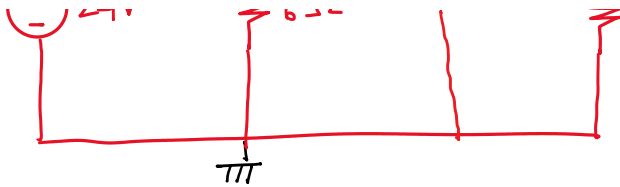


Apply Node Analysis:

$$\frac{V_1 - 24}{4} + \frac{V_1}{6} + \frac{V_1}{3} = 0$$

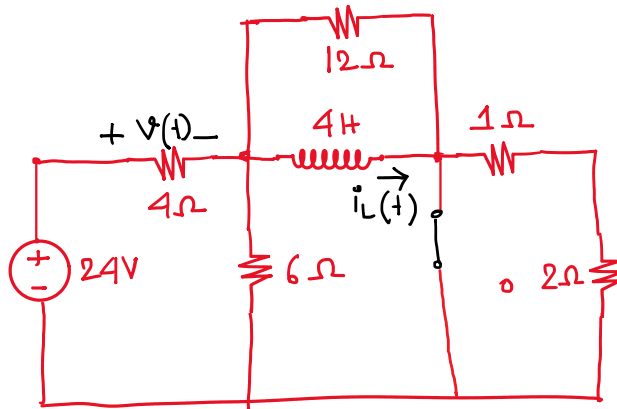
$$0.402 V_1 = 6 \Rightarrow V_1 = \frac{6}{0.402} = 14.89$$

∴ V_1



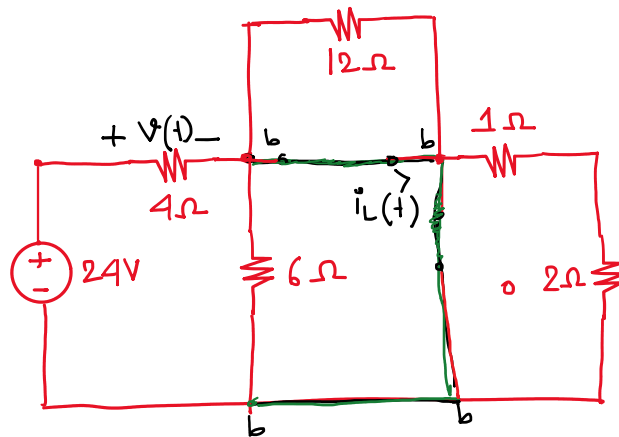
$$0.402 V_1 = 6 \Rightarrow V_1 = \frac{6}{0.402} = 14.89$$

$$i_L(0^-) = \frac{V_1}{3} = \frac{14.89}{3} = 4.96 \text{ A}$$



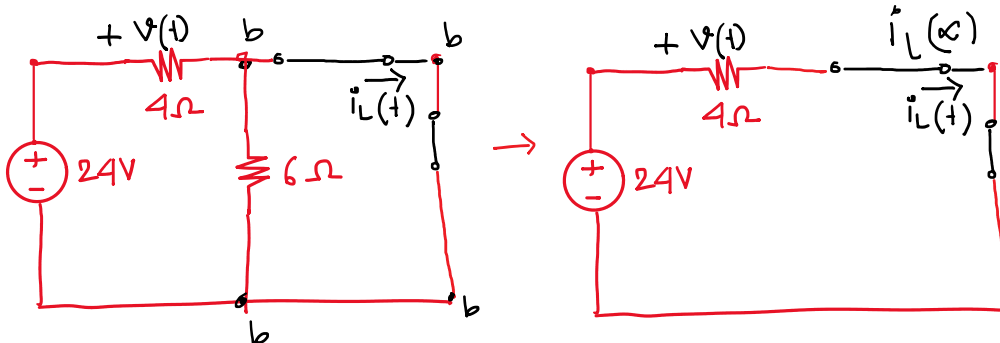
$i_L(\infty)$, $t \geq 0$ Switch closed

$t = \infty$ inductor s.c.k.r. to DC signal

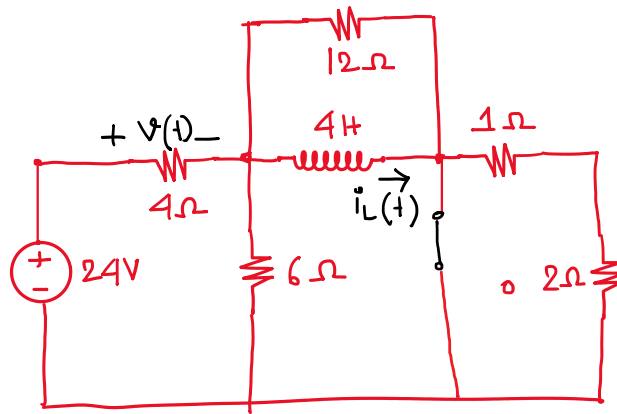


b: Same node.

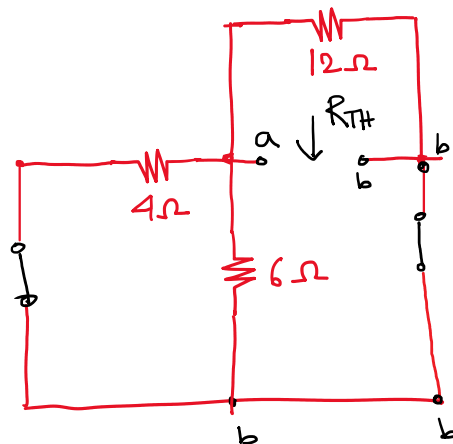
12Ω shorted, also 3Ω shorted, 6Ω shorted



$$i_L(\infty) = \frac{24}{4} = 6 \text{ A}$$



Remove L



betⁿ a & b: $12\Omega, 4\Omega, 6\Omega$ are connect.

$$R_{eq} = R_{TH} = 12 \parallel (4 \parallel 6) = 2\Omega$$

$$\tau = \frac{L}{R_{eq}} = \frac{4}{2} = 2 \text{ sec}$$

$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)] e^{-t/\tau}$$

$$i_L(t) = 6 + [4.96 - 6] e^{-t/2} = 6 - 1.04 e^{-0.5t}$$

